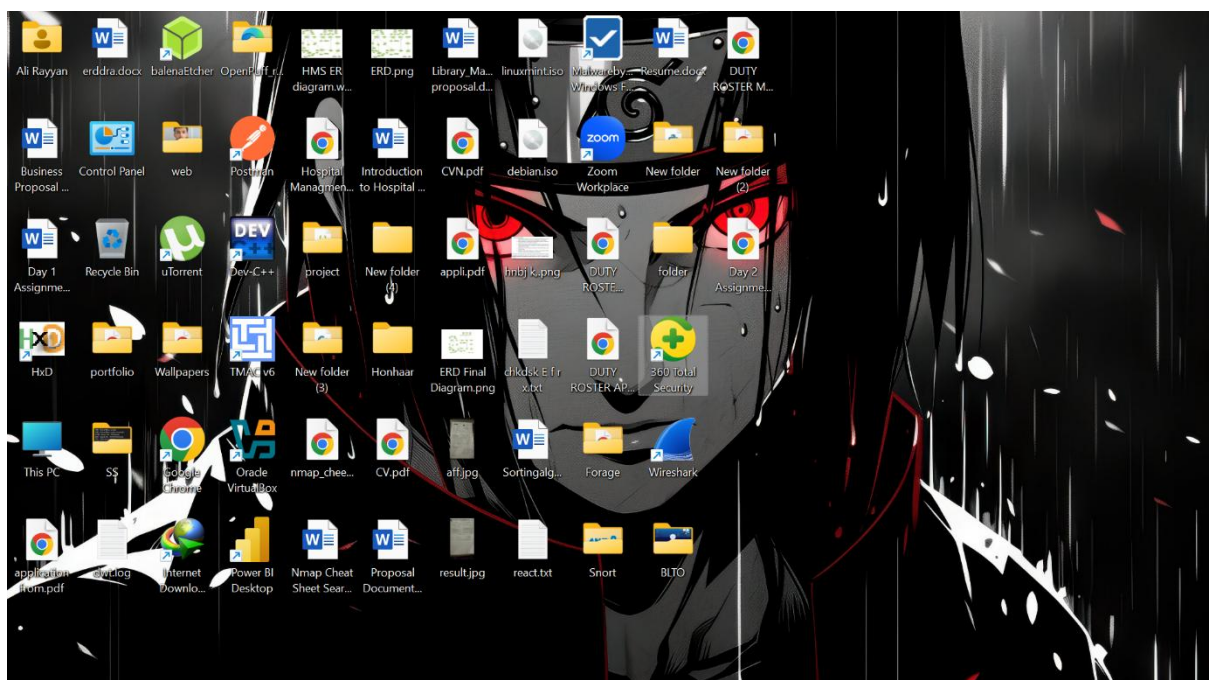
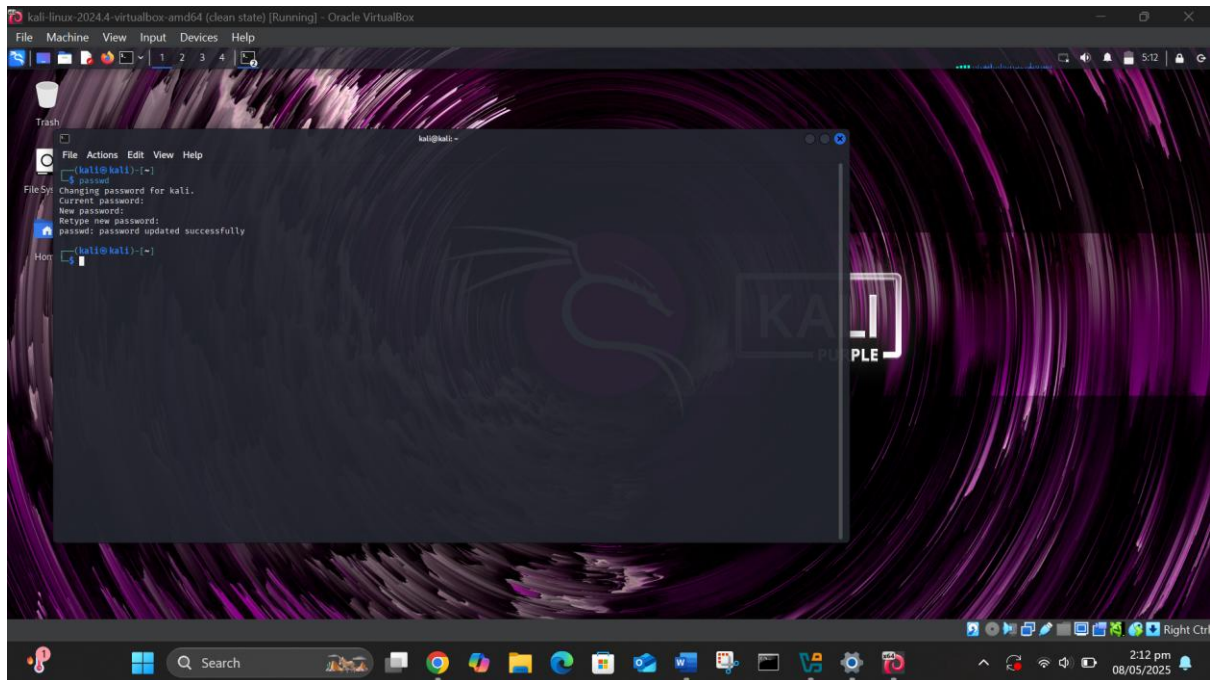


Day 3 Assignment Manual – Linux Essentials & Hacking

Lab Setup

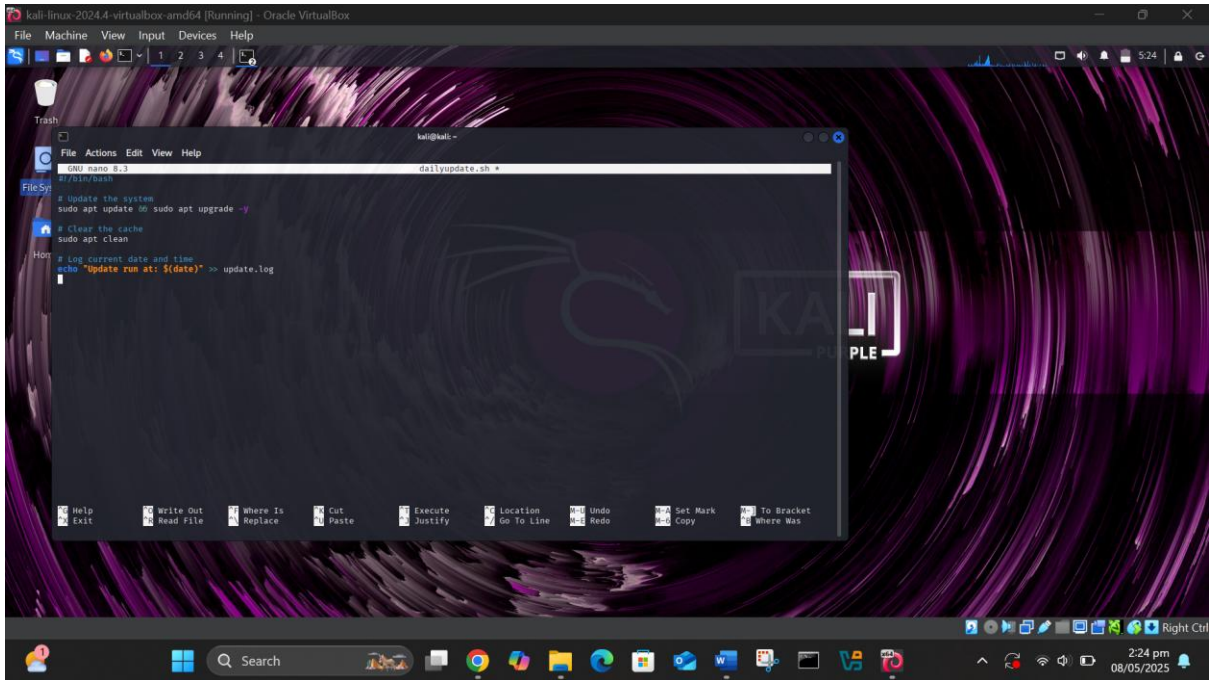
QUESTION #01

- Kali Linux Installation + Setup (*Practical Onboarding Task*)



QUESTION #02

- Create a System Maintenance Script (*System Admin Skill*)



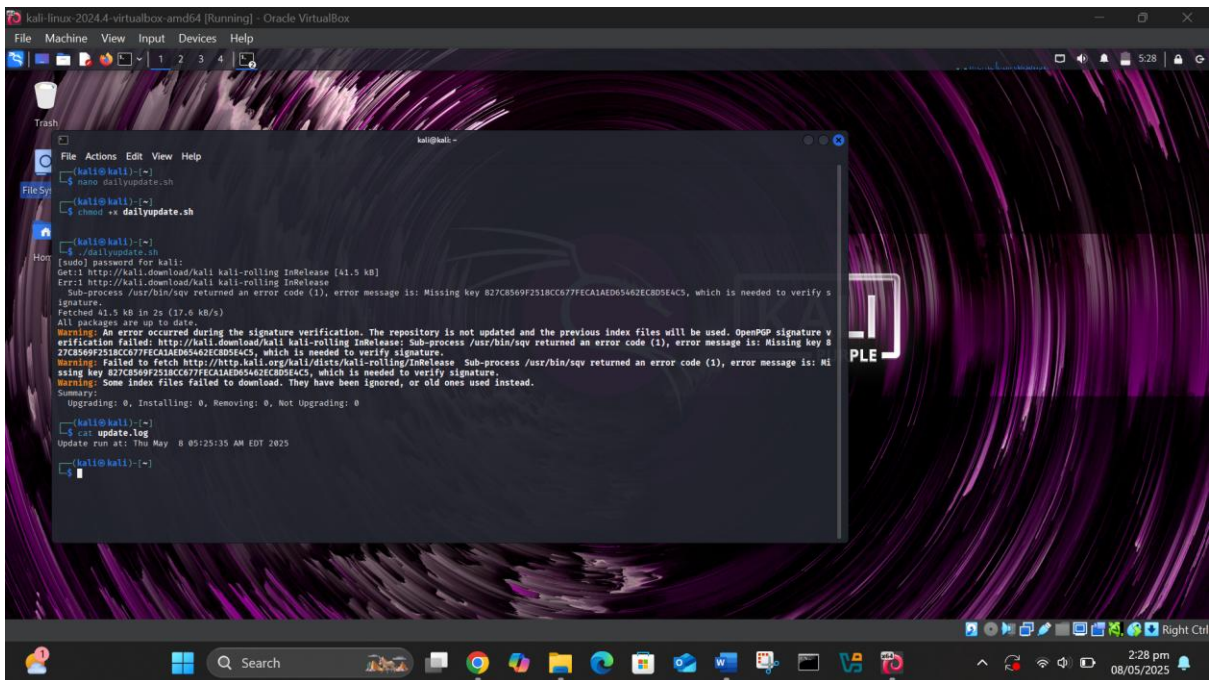
The screenshot shows a Kali Linux virtual machine interface. A terminal window is open, displaying the creation of a script named `dailyupdate.sh`. The script content is as follows:

```
#!/bin/bash

# Update the system
sudo apt update && sudo apt upgrade -y

# Clear the cache
sudo apt clean

# Log current date and time
echo "Update run at: $(date)" >> update.log
```



The screenshot shows the same Kali Linux virtual machine interface. The terminal window now shows the execution of the `dailyupdate.sh` script. The output indicates that the system update failed due to missing GPG keys. The terminal output is as follows:

```
(kali@kali)-[~]
└─$ ./dailyupdate.sh
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Err:1 http://kali.download/kali kali-rolling InRelease
  Sub-process /usr/bin/sq returned an error code (1), error message is: Missing key 827C8569F2518CC677FCA1AE065462EC8D5E4C3, which is needed to verify a signature.
Fetched 41.5 kB in 2s (17.6 kB/s)
All packages are up to date.
Warning: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. OpenPGP signature verification failed: http://kali.download/kali kali-rolling InRelease: Sub-process /usr/bin/sq returned an error code (1), error message is: Missing key 827C8569F2518CC677FCA1AE065462EC8D5E4C3, which is needed to verify signature.
Warning: Failed to fetch http://kali.org/kali/dists/kali-rolling/InRelease Sub-process /usr/bin/sq returned an error code (1), error message is: Missing key 827C8569F2518CC677FCA1AE065462EC8D5E4C3, which is needed to verify signature.
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
(kali@kali)-[~]
└─$ cat update.log
Update run at: Thu May 8 05:25:35 AM EDT 2025
```

QUESTION #03

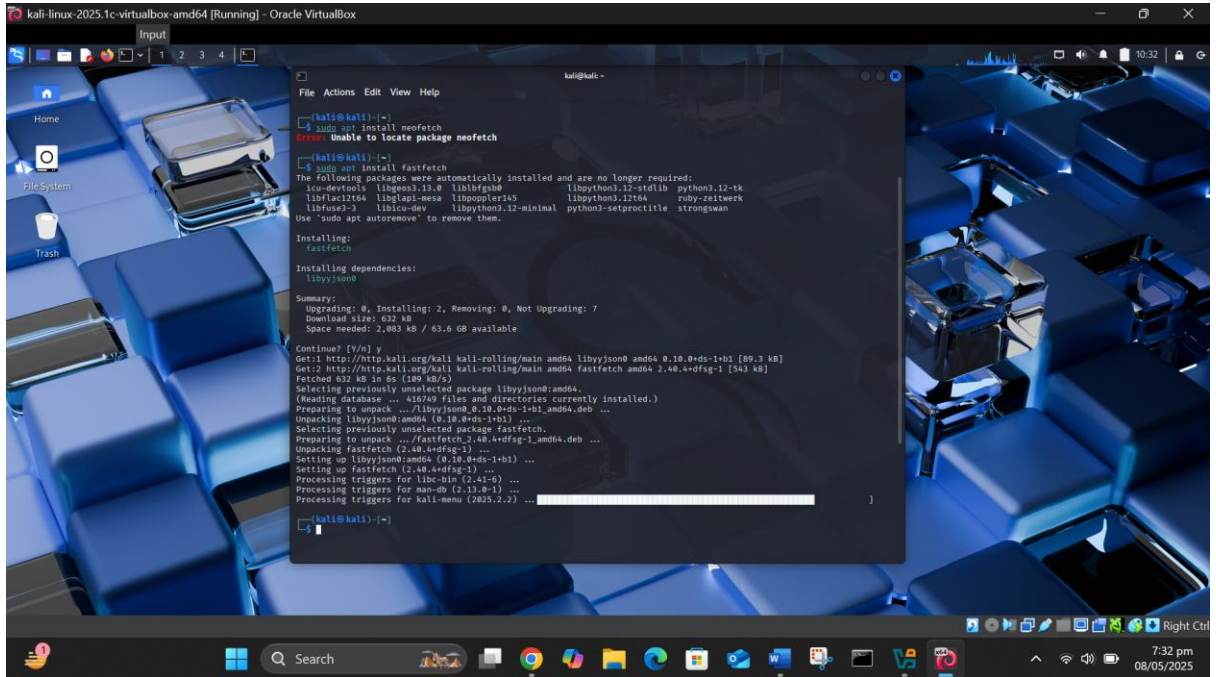
- Explore the File System (*Observation + Internalization*)

Directory Files Found		Purpose/Description
/etc	passwd, hosts, ssh/sshd_config	Contains system-wide configuration files. passwd stores user info, hosts maps hostnames to IPs, and sshd_config configures the SSH server.
/bin	ls, cp, bash	Contains essential binary executables. ls lists files, cp copies files, and bash is the shell program.
/var/log	syslog, auth.log, dpkg.log	Stores system log files. syslog logs general messages, auth.log records authentication attempts, and dpkg.log logs package installations.
/usr/bin	python3, nano, top	Contains user command binaries. python3 is the Python interpreter, nano is a text editor, and top shows system processes.
/home	.bashrc, Documents, Downloads	Stores user directories and files. .bashrc is user shell config, Documents and Downloads store user data.
/boot	vmlinuz, initrd.img, grub	Contains files needed to boot the system. vmlinuz is the Linux kernel, initrd.img initializes RAM disk, and grub is the bootloader folder.
/dev	sda, null, tty	Contains device files. sda is a hard drive, null is a data sink, tty is terminal interfaces.
/proc	cpuinfo, meminfo, uptime	Virtual files providing system info. cpuinfo shows CPU details, meminfo shows RAM usage, uptime shows system uptime.
/lib	ld-linux.so.2, libc.so.6, modules	Contains essential shared libraries for binaries in /bin and /sbin.
/tmp	Temporary files like tmp1234, sess_abc, etc.	Stores temporary data created by programs, deleted on reboot.

Question #04

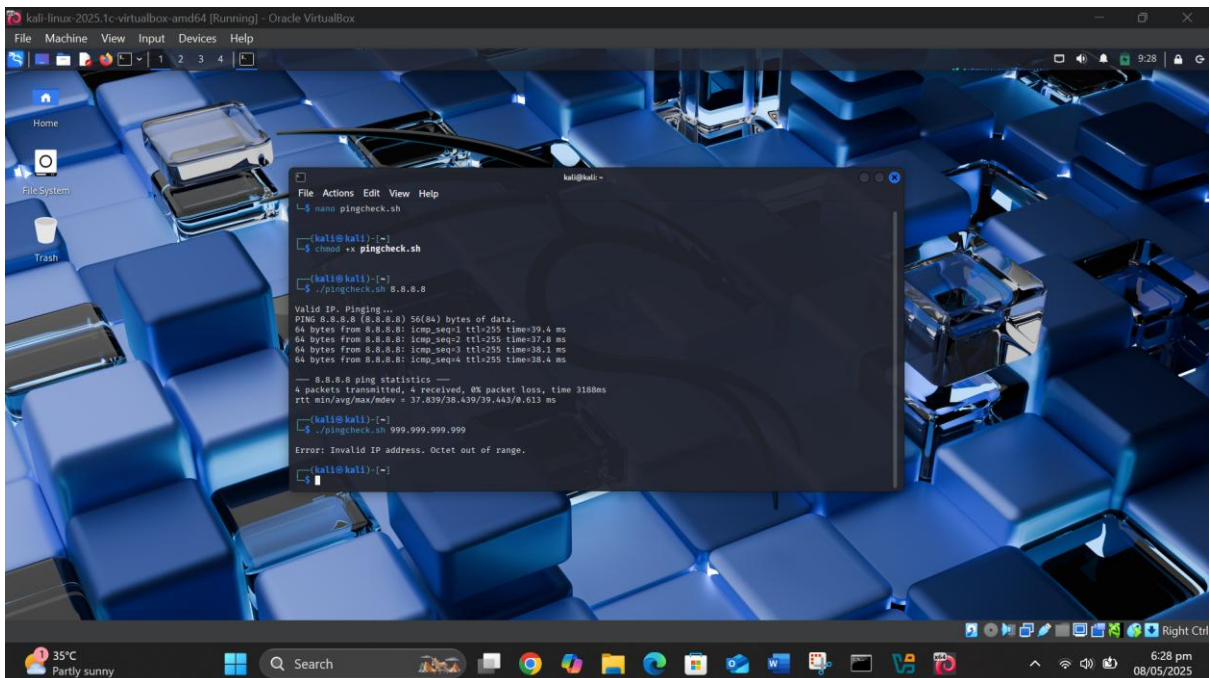
- **Installing a new tool**
- It's commonly used by **system administrators, developers, and Linux enthusiasts** to quickly check system specs or showcase their setup in screenshots, especially for documentation or community sharing. It's also

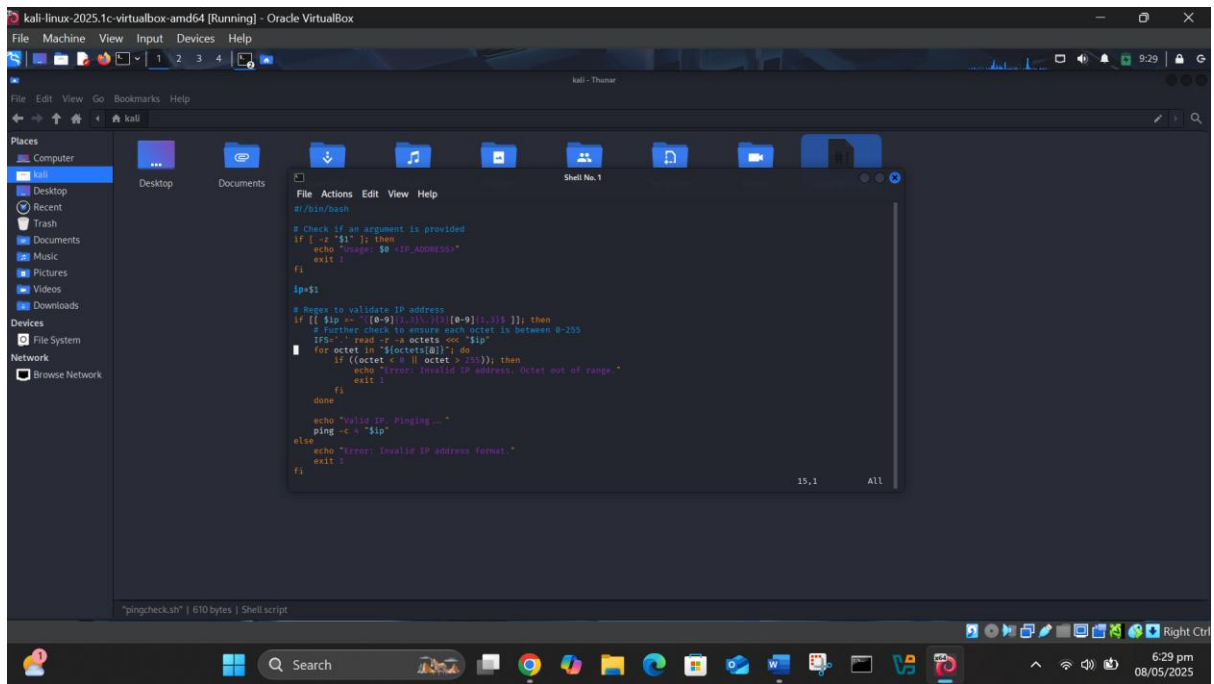
useful in **custom scripts** or **monitoring dashboards** where minimal output speed is crucial.



QUESTION #05

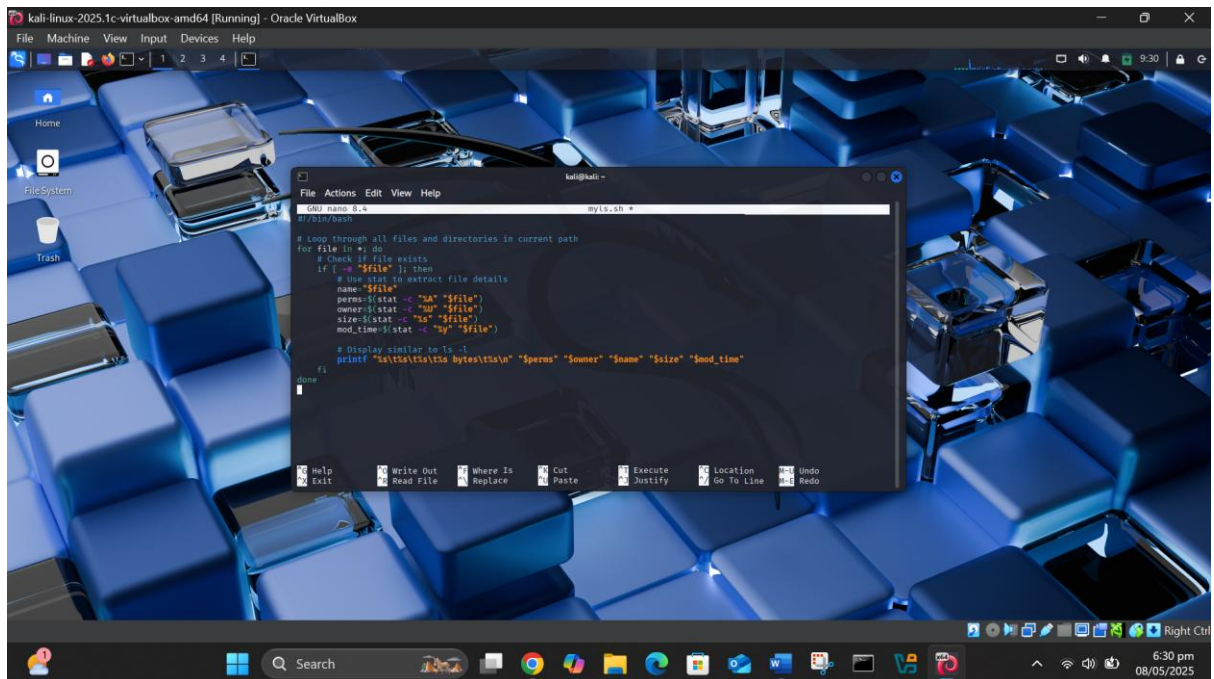
- **Shell Script with IP Validation (*Regex & Conditional Logic*)**

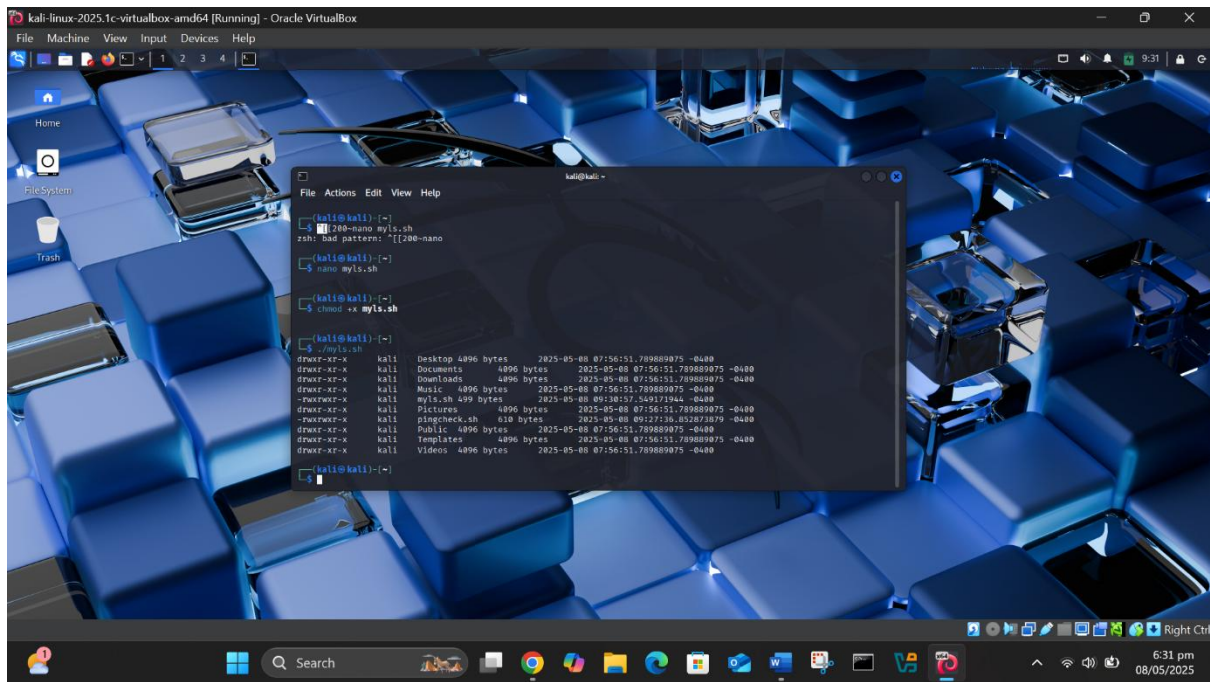




QUESTION #06

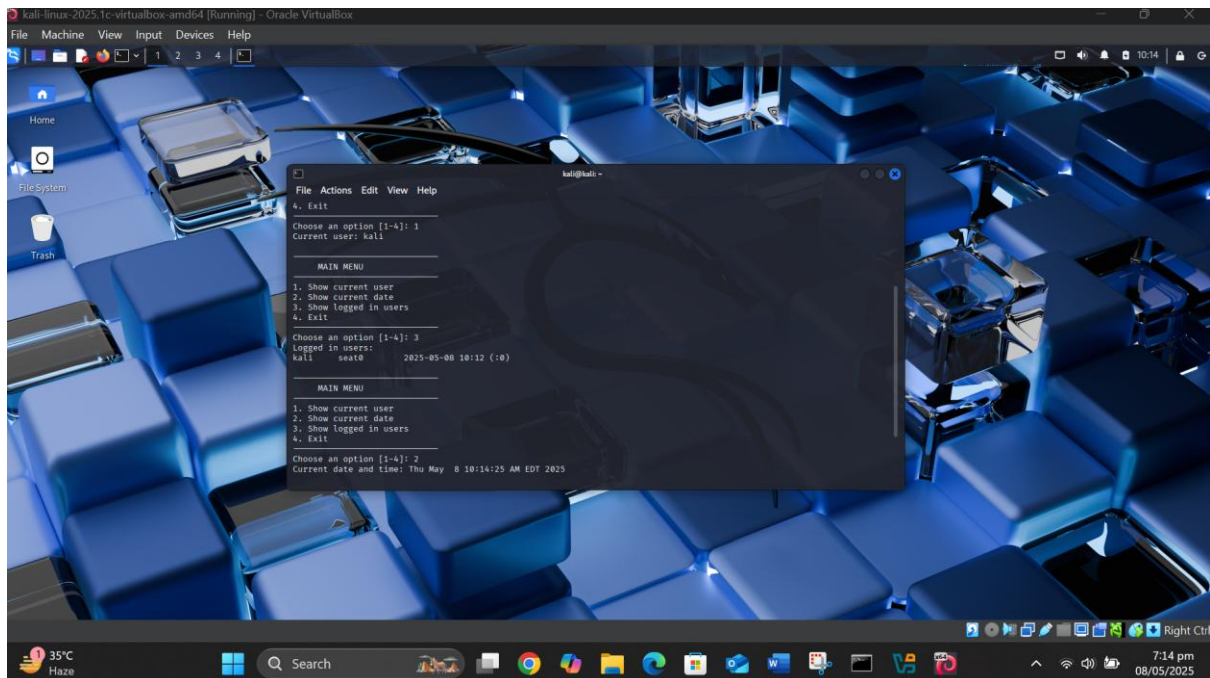
- **Custom Is Clone with Bash** (*Command-Line Mastery*)





QUESTION #07

- **Bash Script With Menu System (*UX Logic in Shell*)**



Summary: How `chmod` and `chown` were used

<code>fileA.txt</code>	<code>chmod 400 fileA.txt</code>	Only user can read the file
<code>fileB.txt</code>	<code>chmod 440 fileB.txt</code>	Group can read but not write
<code>fileC.txt</code>	<code>sudo chown root:root fileC.txt</code> + <code>chmod 700</code>	Only root can access the file